The Likelihood of a Computed Dynamic Model: Proposition 2 of Fernández-Villaverde, Rubio-Ramírez and Santos, Corrected

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Abstract

Proposition 2 of Fernández-Villaverde et al. [2006] bounds the gap between the exact and the computed likelihood of a dynamic model by a term linear in the sample length, $TB\chi\delta$, with constants that do not depend on T. Ackerberg et al. [2009], following Geweke [2007], show by a Gaussian counterexample that this is false. We locate the error in a single step of the proof: a difference of two T-fold products of densities is bounded as if the other $T-1$ factors were absent, so that the "constant" B is really M^{T-1}, where M bounds the density. We then show that the proposition survives, with the same shape, once it is stated for the *logarithm* of the likelihood: under the paper's lower bound on densities, a pathwise version of its contraction condition, and a Lipschitz condition on the log density in the initial state, $|\log L_j(y^T) - \log L(y^T)| \le (T\chi/\xi + K/(1-\alpha))\,\delta$. The dependence on T is sharp. The average log-likelihood error is therefore $O(\delta)$ uniformly in T, which is the quantity Ackerberg, Geweke and Hahn identify as relevant for estimation. The argument is formalised in Lean 4 with Mathlib, including the existence of the coupling of invariant distributions (by Krylov–Bogolyubov), the derivation of the hypotheses from primitives in the case where the measurement shock alone is recovered from the data, and explicit bounds on the resulting posteriors.

1 Setting

Fernández-Villaverde et al. [2006] (FRS) study a state-space model

$$S_t = \varphi(S_{t-1}, W_t; \gamma), \qquad Y_t = g(S_t, V_t; \gamma),$$

with compact supports, a unique invariant distribution μ^* of the state, and computed approximations φ_j, g_j satisfying $\|\varphi_j - \varphi\| \le \delta$ and $\|g_j - g\| \le \delta$ in the sup norm. The shock

*The Lean development is `LeanEconomics/Econometrics/ApproxLikelihood.lean` in `https://github.com/LeanEconomics/LeanEconomics`. It was written mostly by Claude (Anthropic), an AI coding model, working under the direction of Robert Kirkby (Victoria University of Wellington); every result marked with a formal statement was checked by the Lean kernel and uses only Lean's three standard axioms.

W_t is split as (W_{1t}, W_{2t}), where (W_{2t}, V_t) is recovered from the data by inverting the system. Write $\omega = W_1^T$ for the unobserved shock path and

$$p_t(s,\omega) = p(y_t \mid W_1^t = \omega_{\le t},\, S_0 = s,\, y^{t-1};\gamma)$$

for the conditional density of the t-th observation, with q_t its counterpart under (φ_j, g_j). Then

$$L(y^T) = \iint \prod_{t=1}^{T} p_t(s,\omega)\, Q(\mathrm{d}\omega)\, \mu^*(\mathrm{d}s), \qquad L_j(y^T) = \iint \prod_{t=1}^{T} q_t(s,\omega)\, Q(\mathrm{d}\omega)\, \mu_j^*(\mathrm{d}s),$$

where Q is the law of W_1^T and μ_j^* the invariant distribution of the computed chain. FRS's Assumptions 4 and 8 give a lower bound $p_t, q_t \ge \xi > 0$; their Lemma 6 gives $|q_t - p_t| \le \chi\delta$; and their Condition 1 is the mean contraction $\int \|\varphi(S,W) - \varphi(S',W)\|\, \mathrm{d}Q(W) \le \alpha\|S - S'\|$ with $\alpha < 1$.

FRS's Proposition 2. Under Assumptions 1–11 and Condition 1, there are constants B and L such that, for all T,

$$|L_j(y^T) - L(y^T)| < \Big(TB\chi + \frac{L}{1-\alpha}\Big)\delta.$$

2 Where the proof fails

FRS split the gap as $|L_j - L| \le |\int F\,\mathrm{d}\mu_j^* - \int F\,\mathrm{d}\mu^*| + |\int (F_j - F)\,\mathrm{d}\mu_j^*|$, with $F = \int \prod_t p_t\,\mathrm{d}Q$ and $F_j = \int \prod_t q_t\,\mathrm{d}Q$. Their bound (14) on the second term is $TB\chi\delta$. But the telescoping identity

$$\prod_{t=1}^{T} q_t - \prod_{t=1}^{T} p_t = \sum_{t=1}^{T} \Big(\prod_{s<t} q_s\Big)(q_t - p_t)\Big(\prod_{s>t} p_s\Big)$$

places $T-1$ density factors beside each error $q_t - p_t$. With M a bound on the densities the honest bound is $TM^{T-1}\chi\delta$, and M^{T-1} is not a constant: it grows exponentially when $M > 1$ and vanishes exponentially when $M < 1$. This is exactly the factor $(\sigma\sqrt{2\pi})^{-(T-1)}$ in Ackerberg, Geweke and Hahn's sharp bound for the Gaussian location model. The step cannot be repaired with any T-free constant:

Proposition 1. *Let $M > 1$ and $\varepsilon > 0$. There is no B such that $|(M+\varepsilon)^T - M^T| \le TB\varepsilon$ for every $T \in \mathbb{N}$.*

Formal statement: `not_exists_linear_bound_prod_sub`.

Proof. By Bernoulli's inequality $(M+\varepsilon)^{n+1} - M^{n+1} \ge (n+1)\varepsilon M^n$, so the bound would force $M^n \le B$ for all n. □

Two further steps let constants depend on T without comment. The Lipschitz constant of F in S_0, used in (12), is "different for each t, but since t is finite, we can set a global L"; the proposition is then stated "for all T". And Lemma 6's χ is asserted for all t "since the

model is stationary", although the recovered (v_t, s_t, w_{2t}) depend on a t-fold composition of the transition. We return to both in Section 4.

3 The corrected proposition

The accumulation of error that FRS describe is real, but it is *multiplicative*: it holds for the log likelihood, or equivalently the likelihood ratio, and not for the level.

Hypothesis 1 (Uniform per-period error)**.** $\xi \le p_t(s,\omega)$, $\xi \le q_t(s,\omega)$ and $|q_t(s,\omega) - p_t(s,\omega)| \le \varepsilon$ for all t, s, ω, with $\xi > 0$. In FRS's notation $\varepsilon = \chi\delta$.

Hypothesis 2 (Log-Lipschitz in the initial state)**.** There are $\ell_t \ge 0$ with $p_t(s',\omega) \le e^{\ell_t d(s,s')} p_t(s,\omega)$ for all s, s', ω. Write $K = \sum_{t=1}^{T} \ell_t$. Only the *exact* densities need this.

Hypothesis 3 (Coupled initial states)**.** There is a probability measure P on a space X and maps $a, b : X \to S$ whose laws are μ^* and μ_j^*, with $d(a(x), b(x)) \le \rho$ for P-almost every x.

Proposition 2 (Proposition 2′)**.** *Under Hypotheses 1–3, and integrability of the likelihood integrands,*

$$e^{-D} L(y^T) \le L_j(y^T) \le e^{D} L(y^T), \qquad D = \frac{T\varepsilon}{\xi} + K\rho,$$

equivalently $|\log L_j(y^T) - \log L(y^T)| \le D$.

Formal statement: `likelihood_le_exp_mul, abs_log_likelihood_sub_le`.

Proof. Step 1 (one factor). If $\xi \le p$ and $|q - p| \le \varepsilon$ then $q \le p + \varepsilon \le p(1 + \varepsilon/\xi) \le e^{\varepsilon/\xi} p$.
Formal statement: `le_exp_div_mul`.

Step 2 (the product; this replaces FRS's (14)). Multiplying Step 1 over t, $\prod_t q_t \le e^{T\varepsilon/\xi} \prod_t p_t$ pointwise in (s, ω).
Formal statement: `prod_le_exp_card_mul_prod`.

Step 3 (moving the initial state). Multiplying Hypothesis 2 over t, $\prod_t p_t(s',\omega) \le e^{Kd(s,s')} \prod_t p_t(s,\omega)$.

Step 4 (integrating; this replaces FRS's (12)). Combining Steps 2 and 3 and integrating over ω, $\int \prod_t q_t(s',\cdot)\,\mathrm{d}Q \le e^{T\varepsilon/\xi + Kd(s,s')} \int \prod_t p_t(s,\cdot)\,\mathrm{d}Q$ for all s, s'. Put $s = a(x)$, $s' = b(x)$, use $d(a(x), b(x)) \le \rho$ almost surely, and integrate against P: $L_j \le e^D L$.
Formal statement: `lik_le_exp_mul_lik`.

For the reverse inequality, take the two moves in the other order: first move the initial state from $a(x)$ to $b(x)$ *within the exact model* (Step 3), then switch to the computed model at $b(x)$ (Step 2, with the roles of p and q exchanged, which uses $\xi \le q_t$). This is why Hypothesis 2 is needed only for the exact densities.
Formal statement: `lik_le_exp_mul_lik_rev`.
Since the densities are at least ξ and Q, P are probability measures, $L, L_j \ge \xi^T > 0$, and taking logs gives the second form. □

Remark 3 (Sharpness in T). With every exact density equal to M and every computed one to $M + \varepsilon$, $\log L_j - \log L = T \log(1 + \varepsilon/M)$ exactly. The bound cannot grow more slowly

than linearly in T. In Ackerberg, Geweke and Hahn's location model restricted to a compact support, taking every y_t at the same point $\gamma + c$ gives $\log L_j - \log L = T(c\delta - \delta^2/2)/\sigma^2$, again linear in T.

Formal statement: `log_prod_div_prod_const`.

4 The hypotheses from primitives

Hypothesis 1 is FRS's Assumptions 4 and 8 and the conclusion of their Lemma 6, stated uniformly in t. Hypotheses 2 and 3 replace their Condition 1 and the Santos–Peralta-Alva bound used in (12). We derive all three from primitives on the model, with proofs in the appendices. Two of FRS's assumptions are strengthened.

- *Pathwise contraction.* Condition 1 is strengthened from contraction in mean to contraction for every shock: $d(\varphi(s,w),\varphi(s',w)) \le \alpha\, d(s,s')$ for all w.
- *No W_2.* We take $\dim V_t = \dim Y_t$, so that the measurement shock alone is recovered from the data. The state path $s_{t+1} = \varphi(s_t, \omega_t)$ is then a function of the initial state and the shock path only, and the conditional density of the t-th observation is $f_t(s_t)$ for a function f_t of the current state: the density of V at $g^{-1}(y_t; s)$ times the Jacobian.

Theorem 4 (Proposition 2′ from primitives)**.** *Let S be a compact metric space and W a measurable space carrying the shock law ν. Let $\varphi, \varphi_j : S \times W \to S$ be jointly measurable and continuous in the state, with φ a pathwise α-contraction ($0 \le \alpha < 1$) and $d(\varphi_j(s,w),\varphi(s,w)) \le \delta$. Suppose each chain has a unique invariant distribution, μ^* and μ_j^*. Let the measurable per-state densities $f_t, f_{j,t}$ take values in $[\xi, M]$ with $\xi > 0$, satisfy $|f_{j,t} - f_t| \le \eta$, and let each f_t be c-Lipschitz. Then for every sample length T and every law Q of the shock path,*

$$|\log L_j(y^T) - \log L(y^T)| \le \frac{T}{\xi}\Big(\eta + \frac{c\,\delta}{1-\alpha}\Big) + \frac{c\,\alpha}{\xi}\frac{\delta}{(1-\alpha)^2}.$$

Formal statement: `abs_log_likelihood_sub_le_of_primitives`.

Measurability and integrability of all the integrands are derived, not assumed. The three ingredients:

- *The coupling* (Appendix A). Drive both chains with the same shocks. The band $\{(s,s') : d(s',s) \le \delta/(1-\alpha)\}$ is compact and invariant under the joint chain, so by Krylov–Bogolyubov the joint chain has an invariant probability measure on it. Its marginals are invariant for each chain, hence equal to μ^* and μ_j^*.
- *Lemma 6 uniformly in t* (Appendix B). The computed and exact state paths from a common initial state stay within $\delta/(1-\alpha)$ of each other, so $|q_t - p_t| \le \eta + c\delta/(1-\alpha)$ for every t.
- *Lipschitz in the initial state* (Appendix B). The exact chain forgets its initial state at rate α^t, so $\ell_t = (c\alpha/\xi)\alpha^t$ and $K \le c\alpha/(\xi(1-\alpha))$ for every T.

When W_2 is not empty, the recovered shocks depend on the data and on the whole past path, and Hypotheses 1 and 2 become a stability condition on the inverted filter. It is not implied by FRS's Condition 1, and we have not found a primitive condition that delivers it. The nearest results are on forgetting of the initial condition in hidden Markov models [Bickel et al., 1998, Douc et al., 2004] and on uniform-in-time approximation of nonlinear filters [Le Gland and Oudjane, 2004]. They assume a transition density bounded above and below, which does not fit FRS's setting, where the state moves by a deterministic map of the shocks.

The general form with geometric decay, without the no-W_2 structure, is:

Corollary 5 (Constants uniform in T)**.** *Under Hypotheses 1–3, if $\ell_t \le c\alpha^t$ and $\rho = \delta/(1-\alpha)$, then for every T*

$$|\log L_j(y^T) - \log L(y^T)| \le \frac{T\varepsilon}{\xi} + \frac{c\,\delta}{(1-\alpha)^2}, \qquad \frac{1}{T}|\log L_j(y^T) - \log L(y^T)| \le \frac{\varepsilon}{\xi} + \frac{c\,\delta}{T(1-\alpha)^2}.$$

Formal statement: `abs_log_likelihood_sub_le_of_geometric`.

This is FRS's bound, $T \cdot \text{const} \cdot \delta + \text{const} \cdot \delta/(1-\alpha)$, with the level replaced by the log.

5 Posteriors

Suppose Proposition 2′ holds at every parameter value γ with a common D, so that $e^{-D}L(\gamma) \le L_j(\gamma) \le e^{D}L(\gamma)$ for all γ. Write $\pi(\gamma \mid y^T) = L(\gamma)/Z$ and $\pi_j(\gamma \mid y^T) = L_j(\gamma)/Z_j$ for the posterior densities with respect to a prior π_0, with $Z = \int L \,\mathrm{d}\pi_0$ and $Z_j = \int L_j \,\mathrm{d}\pi_0$.

Proposition 6 (Posteriors)**.** *If $L, L_j > 0$ are π_0-integrable and $Z > 0$, then*

(i) $Z_j \le e^{D} Z$ *and* $Z \le e^{D} Z_j$*;*

(ii) $L_j(\gamma')/L_j(\gamma) \le e^{2D}\, L(\gamma')/L(\gamma)$ *for all γ, γ', so Metropolis–Hastings acceptance ratios are off by at most a factor e^{2D};*

(iii) $\pi_j(\gamma \mid y^T) \le e^{2D}\, \pi(\gamma \mid y^T)$*, and symmetrically;*

(iv) $\int |\pi_j - \pi| \,\mathrm{d}\pi_0 \le e^{2D} - 1$*, so the total variation distance between the posteriors is at most $(e^{2D}-1)/2$;*

(v) for measurable h with $|h| \le B$, the posterior expectations satisfy $|\mathbb{E}_j[h \mid y^T] - \mathbb{E}[h \mid y^T]| \le B\,(e^{2D}-1)$.

Formal statement: `evidence_le_exp_mul, ratio_le_exp_mul_ratio, postDens_le_exp_mul, integral_abs_postDens_sub_le, abs_integral_postDens_sub_le`.

Part (v) replaces FRS's Corollary 4, which asserts convergence $E_j(h \mid y^T) \to E(h \mid y^T)$ without a rate, with an explicit bound. The closest result in the literature is the approximation theorem of Dashti and Stuart [2017, Theorem 4.9 of arXiv:1302.6989v4], building on Cotter et al. [2010]: if the negative log-likelihoods satisfy $|\Phi - \Phi_N| \le M_2(\|u\|)\,\psi(N)$, then the Hellinger distance between the posteriors is at most $C\psi(N)$ for N large. Proposition 2′ supplies exactly that hypothesis, with $\Phi = -\log L$ and $\psi = \delta$ (and M_2 proportional to T).

Proposition 6 is the version with explicit constants that holds for every δ; the proofs are in Appendix C.

6 What survives of FRS's conclusions

- *Error accumulates with T.* True of the log likelihood, at exactly the linear rate FRS describe. In relative terms $|L_j/L - 1| \le e^D - 1 \approx D$ while D is small, so FRS's message "the error is of order $T\delta$" is correct for the *relative* error of the likelihood. Their statement for the absolute error is false.
- *Likelihood ratios, Metropolis–Hastings, posteriors (FRS Corollaries 1–4).* All hold with explicit constants $e^{\pm 2D}$ (Proposition 6).
- *Estimation.* The average log likelihood $\frac{1}{T}\log L_j$ is within $O(\delta)$ of $\frac{1}{T}\log L$, uniformly in T (Theorem 4, Corollary 5). This is the object Ackerberg et al. [2009] identify as relevant for the maximum likelihood estimator, and it agrees with their conclusion that approximation error of order δ moves the estimator's probability limit by $O(\delta)$, not by more as T grows. A sup bound on the objective alone gives only $O(\sqrt{\delta})$ for the maximiser under quadratic identification; their $O(\delta)$ rate uses closeness of derivatives as well.

What does not survive is FRS's conclusion that linearisation "is due to fail as the sample size grows" because the likelihood levels diverge. The levels of both likelihoods go to zero or infinity like M^T anyway. What grows is the log-likelihood gap, and it grows no faster than the log likelihood itself, so the average stays $O(\delta)$. And the posterior bounds of Proposition 6 degrade as e^{2D} with $D \propto T\delta$: for Bayesian inference FRS's warning stands, in the corrected form that $T\delta$ should be small.

A The coupling of the invariant distributions

Lemma 7 (Krylov–Bogolyubov for a random map)**.** *Let X be a nonempty compact metric space, ν a probability measure on a measurable space W, and $F : X \times W \to X$ jointly measurable with $F(\cdot, w)$ continuous for each w. Then the chain $X_{t+1} = F(X_t, W_{t+1})$, $W_t \sim \nu$ i.i.d., has an invariant probability measure: there is μ with $(\mu \otimes \nu) \circ F^{-1} = \mu$.*

Formal statement: `RandomMap.exists_isInvariant`.

Proof. Write $T\mu = (\mu \otimes \nu) \circ F^{-1}$ and $(Mh)(x) = \int h(F(x,w))\,\nu(\mathrm{d}w)$ for bounded continuous h. By change of variables and Fubini, $\int h\,\mathrm{d}(T\mu) = \int Mh\,\mathrm{d}\mu$ (`integral_push`). By dominated convergence (h is bounded and $F(\cdot, w)$ is continuous), Mh is continuous (`continuous_markovFn`): this is the Feller property, and with the duality it makes T weakly continuous. Start from a Dirac measure μ_0 and form the Cesàro averages $\bar{\mu}_n = \frac{1}{n+1}\sum_{k=0}^{n} T^k\mu_0$. The difference $\int Mh\,\mathrm{d}\bar{\mu}_n - \int h\,\mathrm{d}\bar{\mu}_n$ telescopes to $\frac{1}{n+1}(\int h\,\mathrm{d}T^{n+1}\mu_0 - \int h\,\mathrm{d}\mu_0)$, which is at most $2\|h\|/(n+1)$ (`integral_cesaro_sub`). By Prokhorov's theorem the probability measures on compact X form a compact space, so $(\bar{\mu}_n)$ has a cluster point μ. By weak continuity, $\int Mh\,\mathrm{d}\mu = \int h\,\mathrm{d}\mu$ for every bounded continuous h, i.e. $\int h\,\mathrm{d}(T\mu) = \int h\,\mathrm{d}\mu$, and so $T\mu = \mu$. □

The repository already proved this for the income fluctuation problem (`exists_isStationary` in `Distribution/Stationary`), where the shock is finite; the general version differs only in the duality and the Feller step.

Theorem 8 (The coupling)**.** *Let S be a nonempty compact metric space, φ, φ_j jointly measurable and continuous in the state, φ a pathwise α-contraction, $d(\varphi_j(s,w), \varphi(s,w)) \le \delta$, $\delta \ge 0$. If μ^* is the unique invariant distribution of φ and μ_j^* the unique invariant distribution of φ_j, there is a probability measure P on the band $B_\rho = \{(s,s') : d(s',s) \le \rho\}$, $\rho = \delta/(1-\alpha)$, whose first and second marginals are μ^* and μ_j^*.*

Formal statement: `exists_coupling`.

Proof. B_ρ is closed in the compact space $S \times S$, so compact, and nonempty since it contains the diagonal. Define the joint map $G((s,s'),w) = (\varphi(s,w), \varphi_j(s',w))$. It maps B_ρ into itself: $d(\varphi_j(s',w), \varphi(s,w)) \le d(\varphi_j(s',w), \varphi(s',w)) + d(\varphi(s',w), \varphi(s,w)) \le \delta + \alpha\rho = \rho$ (`dist_step_le_of_le`). It is jointly measurable and continuous in the state pair (`jointChain`). By Lemma 7 there is an invariant probability measure P on B_ρ. Taking the first marginal commutes with one step of the chain, because the first coordinate of G is φ of the first coordinate: $(P_1 \otimes \nu) \circ \varphi^{-1} = ((P \otimes \nu) \circ G^{-1})_1 = P_1$ (`push_map_marginal`). So P_1 is invariant for φ, and $P_1 = \mu^*$ by uniqueness. The same argument gives $P_2 = \mu_j^*$. □

The theorem needs uniqueness of the computed chain's invariant distribution, which FRS do not assume. Without it the conclusion holds for *some* invariant distribution of the computed chain, the one the coupling selects.

B The densities from primitives

Let $s_t(s_0, \omega)$ be the exact state path (s_0 given, $s_{t+1} = \varphi(s_t, \omega_t)$) and $s_{j,t}$ the computed one (`statePath`). The likelihood's factors are $p_t(s_0,\omega) = f_t(s_{t+1}(s_0,\omega))$ and $q_t = f_{j,t}(s_{j,t+1})$ (`densPath`).

Lemma 9 (Paths)**.** *For all s_0, s_0', ω, t: (a) $d(s_{j,t}(s_0,\omega), s_t(s_0,\omega)) \le \delta/(1-\alpha)$; (b) $d(s_t(s_0,\omega), s_t(s_0',\omega)) \le \alpha^t d(s_0, s_0')$.*

Formal statement: `dist_statePath_le, dist_statePath_exact_le`.

Proof. (a) With $d_t = d(s_{j,t}, s_t)$, one step gives $d_{t+1} \le \alpha d_t + \delta$ (`dist_step_le`), so by induction $d_t \le \alpha^t d_0 + \delta/(1-\alpha)$ (`dist_iterate_le`), and $d_0 = 0$. (b) Induction on t using the contraction. □

Lemma 10 (Lemma 6, uniformly in t)**.** $|q_t(s_0,\omega) - p_t(s_0,\omega)| \le \eta + c\,\delta/(1-\alpha)$ *for all* t, s_0, ω.

Formal statement: `abs_densPath_sub_le`.

Proof. With $x = s_{t+1}$ and $y = s_{j,t+1}$, $|f_{j,t}(y) - f_t(x)| \le |f_{j,t}(y) - f_t(y)| + |f_t(y) - f_t(x)| \le \eta + c\,d(y,x)$, and $d(y,x) \le \delta/(1-\alpha)$ by Lemma 9(a). □

Lemma 11 (Log-Lipschitz in the initial state)**.** $p_t(s_0', \omega) \le \exp\big((c\alpha/\xi)\,\alpha^t\, d(s_0, s_0')\big)\, p_t(s_0, \omega)$.

Formal statement: `densPath_le_exp_mul`.

Proof. With $x = s_{t+1}(s_0, \omega)$ and $x' = s_{t+1}(s_0', \omega)$, $|f_t(x') - f_t(x)| \le c\, d(x, x') \le c\, \alpha^{t+1} d(s_0, s_0')$ by Lemma 9(b). Since $f_t(x) \ge \xi$, the one-factor step of Proposition 2 turns this additive bound into the multiplicative bound $f_t(x') \le e^{c\alpha^{t+1} d(s_0, s_0')/\xi} f_t(x)$. □

Proof of Theorem 4. Theorem 8 gives the coupling P on the band, with $\rho = \delta/(1-\alpha)$. Lemma 10 is Hypothesis 1 with $\varepsilon = \eta + c\delta/(1-\alpha)$. Lemma 11 is Hypothesis 2 with $\ell_t = (c\alpha/\xi)\alpha^t$, and $\sum_{t<T} \alpha^t \le 1/(1-\alpha)$ (`sum_le_of_geometric`). Corollary 5 then bounds the log-likelihood gap for the likelihoods written against P. Those equal the likelihoods against μ^* and μ_j^* by change of variables along the marginal maps.

Measurability is derived. The state path is jointly measurable in (s_0, ω) by induction (`measurable_statePath`), so the conditional likelihood $\prod_t f_t(s_{t+1})$ is jointly measurable. By Fubini its integral over ω is measurable in s_0 (`integrable_densPath`). The bounds $0 \le \prod_t f_t \le M^T$ give integrability against every probability measure. □

C Proofs for the posteriors

Proof of Proposition 6. (i) Integrate $L_j \le e^D L$ against the prior. (ii) $L_j(\gamma') \le e^D L(\gamma')$ and $1/L_j(\gamma) \le e^D/L(\gamma)$. (iii) Divide $L_j(\gamma) \le e^D L(\gamma)$ by $Z_j \ge e^{-D} Z$. (iv) Write $a = \pi_j(\gamma \mid y^T)$, $b = \pi(\gamma \mid y^T)$, both positive. By (iii) applied both ways, $a \le e^{2D} b$ and $b \le e^{2D} a$. Hence $a - b \le (e^{2D} - 1)\, b$, and $b - a \le b - e^{-2D} b = (1 - e^{-2D})\, b \le (e^{2D} - 1)\, b$, so $|a - b| \le (e^{2D} - 1)\, b$ (`abs_postDens_sub_le`). Integrate, using $\int b \,\mathrm{d}\pi_0 = 1$. (v) $|\int h(a-b)\,\mathrm{d}\pi_0| \le B \int |a-b|\,\mathrm{d}\pi_0$, then (iv). □

D The formal development

Result	Lean name
Econometrics/ApproxLikelihood	
One factor, additive to multiplicative	`le_exp_div_mul`
Corrected product step (14)	`prod_le_exp_card_mul_prod`
FRS's step (14) cannot hold	`not_exists_linear_bound_prod_sub`
Sharpness in T	`log_prod_div_prod_const`
Coupled chain: one step, band, path	`dist_step_le_of_le`, `dist_iterate_le`
Proposition 2′, both directions	`lik_le_exp_mul_lik`, `..._rev`
Proposition 2′, log form	`abs_log_likelihood_sub_le`
Corollary 5	`abs_log_likelihood_sub_le_of_geometric`
Distribution/RandomMap	
Lemma 7 (Krylov–Bogolyubov)	`RandomMap.exists_isInvariant`
Econometrics/ApproxLikelihoodModel	
Theorem 8 (coupling)	`exists_coupling`
Lemma 9	`dist_statePath_le`, `dist_statePath_exact_le`
Lemma 10	`abs_densPath_sub_le`
Lemma 11	`densPath_le_exp_mul`
Theorem 4	`abs_log_likelihood_sub_le_of_primitives`
Econometrics/ApproxPosterior	
Proposition 6 (i)–(iii)	`evidence_le_exp_mul`, `postDens_le_exp_mul`
Proposition 6 (iv)	`integral_abs_postDens_sub_le`
Proposition 6 (v)	`abs_integral_postDens_sub_le`

Paths are relative to `LeanEconomics/`. The library builds without `sorry`, and every result above depends only on Lean's three standard axioms (`propext`, `Classical.choice`, `Quot.sound`).

What remains informal: the reduction of FRS's measurement equation to per-state densities f_t (the density of V at $g^{-1}(y_t; s)$ times the Jacobian) and the bound $\eta = O(\delta)$ on their error at a fixed state, which is FRS's Lemma 6 argument without the time dimension; and the case with W_2 present, discussed in Section 4.

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